

Multi-Valued Dependency (MVD)

A **Multi-Valued Dependency (MVD)** exists when:

One attribute determines **multiple independent values** of another attribute.

$$A \twoheadrightarrow B$$

(Read: A multi-determines B)

Thus, for a single value of **A**, there can be **multiple values of B** and those values are **independent of other attributes**

Example

Table: Student

Student	Course	Hobby
Rahul	Math	Cricket
Rahul	Math	Music
Rahul	Science	Cricket
Rahul	Science	Music

- Rahul has multiple **Courses** $\rightarrow$ Math, Science
- Rahul has multiple **Hobbies** $\rightarrow$ Cricket, Music

But, Courses and Hobbies are **independent**

So MVDs are:

- Student $\twoheadrightarrow$ Course
- Student $\twoheadrightarrow$ Hobby

Problem

The table creates **all possible combinations**:

- Math + Cricket
- Math + Music
- Science + Cricket

- Science + Music

This Leads to: Redundancy, Data duplication, Inefficiency

MVD itself is **not wrong**, but the problem occurs when **More than one independent MVD exists in the same table**

Fourth Normal Form (4NF)

A relation is in **Fourth Normal Form (4NF)** if:

1. It should be in BCNF
2. It should **not have any non-trivial Multi-Valued Dependency**

Converting to 4NF (Decomposition)

Original Table (NOT in 4NF)

Student	Course	Hobby
Rahul	Math	Cricket
Rahul	Math	Music
Rahul	Science	Cricket
Rahul	Science	Music

Splitting the table

Table 1: Student_Course

Student	Course
Rahul	Math
Rahul	Science

Table 2: Student_Hobby

Student	Hobby
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Rahul	Cricket
Rahul	Music

Thus, it eliminates:

- No unnecessary combinations
- No redundancy
- Each table has only **one MVD**

Benefits of Decomposition:

1. No repetition of unrelated attribute combinations.

- In a non 4NF table, if two attributes are independently related to a third, their combinations get repeated unnecessarily.
- This leads to a cartesian product effect, lots of rows just to represent all combinations.

2. Each Table Contains Data with a Single Multivalued Dependency

- Every decomposed table focuses on only one multivalued relationship.

Practice Questions

Question 1: Check whether the following table is in 4NF? If not, convert it into 4NF.

Student	Sport	Music
Ravi	Cricket	Guitar
Ravi	Cricket	Piano
Ravi	Football	Guitar
Ravi	Football	Piano
Neha	Tennis	Dance
Neha	Tennis	Singing
Neha	Badminton	Dance
Neha	Badminton	Singing

Question 2: Check whether the following table is in 4NF? If not, convert it into 4NF.

Teacher	Subject	Book
Meena	Math	Book1
Meena	Math	Book2
Meena	Science	Book1
Meena	Science	Book2
Raj	English	Book3
Raj	English	Book4
Raj	History	Book3
Raj	History	Book4

Fifth Normal Form (5NF)

5NF is one of the highest levels of normalization, also known as Project-Join Normal Form (PJNF).

A table is said to be in 5NF if:

- It is already in 4NF
- It cannot be broken down into smaller tables without losing data

In other words, there should be no join dependency left that can cause redundancy.

The main goal of 5NF is to break down a table into the smallest possible pieces while making sure that:

- You can still reconstruct the original data without loss, and
- No unnecessary repetition of data occurs.

Join Dependency (JD)

A Join Dependency (JD) exists when:

A table can be decomposed into **two or more smaller tables**, and the **original table can be reconstructed exactly** by joining them.

i.e. “This table is actually made by joining smaller tables.”

Example

Table: Supplier_Part_Project

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1

There are **3 independent relationships**:

1. Supplier supplies Parts → (Supplier, Part)
2. Part is used in Projects → (Part, Project)

3. Supplier works on Projects $\rightarrow$ (Supplier, Project)

Step 1: Decompose

Split into 3 tables:

1. Supplier_Part

Supplier	Part
S1	P1
S1	P2

2. Part_Project

Part	Project
P1	J1
P1	J2
P2	J1

3. Supplier_Project

Supplier	Project
S1	J1
S1	J2

Step 2: Join back

Now join all three:

Supplier_Part $\bowtie$ Part_Project $\bowtie$ Supplier_Project

We will get the **original table back exactly**

This means: The original table has a **Join Dependency**

Join Dependency means a table can be recreated by joining multiple smaller tables.

Fifth Normal Form (5NF)

A relation is in **5NF** if:

It has **no non-trivial Join Dependency**

Thus, if a table can be split into smaller tables AND still be perfectly reconstructed, then **it should be split**

Example (Apply 5NF)

Original Table:

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1

This table stores:

- Supplier–Part
- Part–Project
- Supplier–Project

All combined → redundancy

Convert to 5NF

Split into 3 tables:

1. Supplier_Part
2. Part_Project
3. Supplier_Project

Now, There is

- No redundancy
- Each table stores one relationship
- Original table can be recreated by join

A relation is in 5NF if it cannot be further decomposed into smaller relations without loss of information, i.e., it has no non-trivial join dependency.

Case 1: (5NF violated)

- Try Decomposition
- Join gives **exact same table, then**
Join Dependency exists and NOT in 5NF

Case 2: (5NF satisfied)

- Try Decomposition
- Join gives **extra/wrong rows, then**
No Join Dependency and Already in 5NF

PRACTICE QUESTIONS

Question 1: Check whether the following table is in 5NF? If not, convert.

Student	Course	Teacher
A	Math	T1
A	Math	T2
A	Science	T1
A	Science	T2

Question 2:

1. Does this table have Multi-Valued Dependency (MVD)?
2. Can it be decomposed into smaller tables and joined back exactly?
3. Is the table in 5NF or not?

Student	Course	Teacher
A	DBMS	X
A	OS	X
B	DBMS	X
B	OS	Y

Question 3:

1. Is the table in 4NF?
2. Can it be decomposed further (5NF check)?
3. What is the highest normal form of this table?

Student	Course	Hobby
A	Math	Cricket
A	Math	Music
A	Science	Cricket
A	Science	Music
B	Math	Cricket
B	Math	Music

Question 4:

1. Check if Multi-Valued Dependency (MVD)?
2. Is the table in 4NF? If not, decompose
3. After 4NF, check for Join Dependency and 5NF.
4. What is the highest normal form?

Employee	Project	Skill	Manager
A	P1	Java	M1
A	P1	Python	M1
A	P2	Java	M2
A	P2	Python	M2
B	P3	C++	M3
B	P3	SQL	M3